



## AI Playbook and Toolkit for Public Health Departments

# APIs, Webhooks, and Event-Driven Workflows

ARC 260

APIs, webhooks, and event-driven workflows are the mechanisms that allow systems to interact. They are especially important for AI because many AI tools need to retrieve data, send notifications, create tasks, call services, or update systems. This module teaches non-developer public health staff enough technical understanding to evaluate integrations, ask stronger questions, and identify safeguards.

|                   |                              |
|-------------------|------------------------------|
| Level             | applied/foundational         |
| Primary track     | Technical Architecture Track |
| Appears in tracks | technical-architecture       |

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## Learning Objectives

- Explain what APIs, webhooks, and event-driven workflows do.
- Distinguish between read-only access, write-back, and triggered actions.
- Identify security, privacy, logging, and access-control requirements for AI integrations.
- Assess where human approval is required before an AI-triggered action occurs.
- Produce an API and event workflow review worksheet.

## Module Overview

APIs, webhooks, and event-driven workflows are the mechanisms that allow systems to interact. They are especially important for AI because many AI tools need to retrieve data, send notifications, create tasks, call services, or update systems. This module teaches non-developer public health staff enough technical understanding to evaluate integrations, ask stronger questions, and identify safeguards.

## Jurisdiction and Agency Policy Note

This national-level module is intended for training and planning purposes. It does not replace legal, privacy, procurement, civil rights, records, or agency policy review. Learners should confirm applicable requirements in their own jurisdiction and organization before implementing AI-supported workflows.

## Why This Topic Matters for Public Health AI

APIs can reduce manual work by allowing systems to exchange data directly. In public health, APIs may retrieve case information, query FHIR resources, send status updates, create tasks, or connect dashboards to source systems. Webhooks can notify a system when a defined event occurs, such as receipt of a new lab result or change in case status. Event-driven workflows can help public health teams respond faster, but they also require strong controls.

The risk increases when AI tools move from reading information to taking action. A read-only AI assistant that summarizes approved documents has a different risk profile than an agent that updates a surveillance record, sends an email, or creates a referral. Public health agencies need to define not only what an integration can do, but what it is allowed to do.

AI integrations also create cybersecurity and privacy obligations. Systems must authenticate users and services, limit permissions, protect credentials, log actions, and handle failures. Prompt injection and malicious inputs can be especially dangerous when an AI tool has access to external systems or sensitive data.

## Core Concepts

**API.** An application programming interface allows one system to request data or services from another system. APIs are useful for structured exchange, but they must be governed through authentication, authorization, scopes, rate limits, monitoring, and logs.

**Webhook.** A webhook sends information automatically when a defined event occurs. In public health, a webhook might trigger a queue update when a new lab result is received or notify staff when a case status changes.

**Event-driven workflow.** An event-driven workflow begins when something specific happens, such as receipt of a report, a threshold being met, a task being completed, or a record being updated. These workflows can improve timeliness but must include error handling and escalation.

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Read-only versus write-back. Read-only access allows a tool to view or retrieve information. Write-back allows a tool to update, create, or delete information. Write-back requires stronger safeguards because it can affect official records and public health actions.

## Public Health Example

An AI tool may be configured to review new environmental health complaints as they arrive and classify them by topic. A webhook could trigger the review when a complaint is submitted. The tool might assign a category, draft a summary, and create a task for human review. This workflow could improve triage, but it must log the AI classification, preserve the original complaint, and allow staff to correct the category.

A higher-risk example is an agentic tool that updates a case management system after interpreting a lab result. Because this action affects a system of record, the workflow should require human approval before write-back. The agency should also test failure scenarios, such as ambiguous results, duplicate records, missing identifiers, and API outages.

## Technical Deep Dive

API review should begin with purpose and permissions. Staff should ask what data the AI tool needs, whether the access is read-only or write-back, which system is authoritative, who or what is authenticated, what authorization scope is granted, and what logs are created. Permissions should follow least privilege, meaning the tool receives only the access required for the approved task.

Event-driven workflows should be designed with failure in mind. The workflow should define what happens if an API call fails, returns incomplete data, times out, receives conflicting information, or triggers duplicate actions. Users should not have to infer whether an automated step occurred. The system should display status, errors, and next steps clearly.

Agentic workflows require the strongest design discipline. When an AI tool can call APIs or use tools, the agency must constrain allowed actions, prohibit sensitive or unauthorized actions, isolate testing environments, monitor outputs, and define rollback. The model itself should not be the only security boundary. Application controls and system permissions should enforce limits.

## Risks, Failure Modes, and Guardrails

Over-permissioned integrations. AI tools may receive broader access than needed. Guardrails include least-privilege scopes, role-based access, periodic access review, and credential management.

Unreviewed write-back. Automated updates can alter official records without adequate oversight. Guardrails include human approval, write-back restrictions, logs, and rollback procedures.

Prompt injection or malicious input. External or user-provided text may manipulate an AI tool connected to systems. Guardrails include input handling, output filtering, tool-use constraints, and separation of untrusted content from instructions.

Poor error handling. Failed API calls may produce incomplete workflows. Guardrails include status indicators, retry rules, exception queues, and staff notification.

## Application to AI-Supported Workflows

Generative AI may use APIs to retrieve source material for summaries. Predictive tools may call services that return risk scores or features. RAG systems may use APIs to retrieve approved documents. Agentic AI may use APIs to create tasks, send messages, or update records, which makes permission design, human approval, and audit logs central to safety.

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## Reflection Questions

Which workflows in your agency are triggered by new data or status changes?

Which AI-supported actions should be read-only?

Which actions would require human approval before write-back?

What logs are needed to reconstruct what happened?

What should happen if an API call fails or returns unexpected data?

## Practical Exercise

Create an API and event workflow review worksheet for one proposed AI integration. Include the triggering event, systems involved, data exchanged, read/write permissions, authentication method, access limits, logging requirements, error handling, human approval points, and escalation pathway.

## Expected Artifact or Evidence

API and event workflow worksheet

Integration risk questions

Access control recommendations

Audit log requirements

## Expected Assignment

- API and event workflow worksheet
- Integration risk questions
- Access control recommendations
- Audit log requirements

## Knowledge Check

- 1. What does an API allow systems to do?
- 2. What is a webhook?
- 3. Why do AI integrations need access controls?
- 4. Which AI action usually requires stronger safeguards?
- 5. What is strong completion evidence for this module?

## References and Resources

- HL7 FHIR API documentation: <https://hl7.org/fhir/>
- ONC interoperability resources
- Agency cybersecurity and access control policies
- OWASP LLM application security resources